

3DEXPERIENCE Message Function

The 3DEXPERIENCE Message Function enables publishing messages from DELMIA Apriso to the 3DEXPERIENCE Platform using an ActiveMQ Message Broker.

The message format is dependent on the selected message.

Prerequisites

To use the 3DEXPERIENCE Message function, a connection to an ActiveMQ broker must be configured in [Configuration Manager](#) .

Additional configuration steps on the 3DEXPERIENCE Platform side are discussed in the ***Integration with 3DEXPERIENCE Platform Guide***.

To Add the 3DEXPERIENCE Message Function

1. Drag the 3DEXPERIENCE Message Function from the [Toolbox](#) to the Function Navigation diagram, or double-click the 3DEXPERIENCE Message icon in the Toolbox.

Properties

General

Name
_3DEXPERIENCEMessage1

Description

Type
3DEXPERIENCE Message

☐ Log Events

3DEXPERIENCE Message

Platform Alias

☐ From PlatformAlias Input

☒ Platform Alias:

☐ Message Topic ☐ Web Service

Message Topic

Refresh Inputs

Name

2. Choose the Message to update the list of available Inputs.

Depending on the selected Platform Alias, the available Topics are different.

Topics available when **Platform Alias** option is selected and the selected alias is deployed on premises:

- ▶ Get Data Collect Plan Row Custom Attributes
- ▶ Get Data Collect Plan Custom Attributes
- ▶ Get Operation Custom Attributes
- ▶ Get Manufacturing Item Custom Attributes
- ▶ Get Process Custom Attributes
- ▶ Get Process Organization
- ▶ Get Resource Implement Links
- ▶ Get Resource Custom Attributes

- ▶ Import IP Protection and Export Control
- ▶ Import Resource
- ▶ Get Process Header
- ▶ Import Process
- ▶ Create Issue
- ▶ Cancel Operation
- ▶ Complete Operation
- ▶ Hold Operation
- ▶ Release Operation
- ▶ Start Operation
- ▶ UnHold Operation
- ▶ Cancel Order
- ▶ Complete Order
- ▶ Hold Order
- ▶ Release Order
- ▶ Start Order
- ▶ Create Post
- ▶ UnHold Order
- ▶ Get Translation
- ▶ Import Translation

Topics available when **Platform Alias** option is selected and the selected alias is deployed in Cloud:

- ▶ Get Data Collect Plan Row Custom Attributes
- ▶ Get Data Collect Plan Custom Attributes
- ▶ Get Operation Custom Attributes
- ▶ Get Manufacturing Item Custom Attributes
- ▶ Get Process Custom Attributes
- ▶ Get Process Organization
- ▶ Get Resource Custom Attributes
- ▶ Get Process Header
- ▶ Import IP Protection and Export Control
- ▶ Import Process

When the **From PlatformAlias Input** option is selected, all topics are available.

Web Service Option

Instead of the Message Topic, you can select Web Service option.

When the Web Service option is selected the Message section is disabled. The **URL** is automatically added as the input, and **Response** and **Code** are added as outputs of the function.

The option allows you to execute a web service call to the API of the platform defined in the selected Platform Alias. The content of the response body of the request is returned in the **Response** output and the response code in the **Code** output.

Production Status Updates

These messages send Production Order and DFC status updates to the 3DEXPERIENCE Platform.

Field	Description
Message	▶ Cancel Operation ▶ Complete Operation ▶ Hold Operation

	▶ Release Operation ▶ Start Operation ▶ UnHold Operation ▶ Cancel Order ▶ Complete Order ▶ Hold Order ▶ Release Order ▶ Start Order ▶ UnHold Order
Refresh Inputs	Refreshes the list of available Inputs (Required and Dynamic) from the Message definition.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
TransactionTime	DateTime	Transaction Time in UTC.
Order3DExperienceID	Char	Order ID in 3DEXPERIENCE Platform: OrderDSID in outgoing json.
Operation3DExperienceID	Char	Operation ID in 3DEXPERIENCE Platform: OperationDSID in outgoing json.
ReasonCode	Char	Reason Code for the status change. ReasonCode in outgoing json.

Dynamic Inputs

The Inputs can be added to the Function by selecting the check boxes on the Dynamic Inputs list.

Name	Data Type	Description
LocalTransactionTime	DateTime	Transaction Time adjusted for local time zone.
OrderNo	Char	Order Number. OrderNo in outgoing json.
OrderType	Int	DELMIA Apriso WIP Order type. The outgoing value is translated into a 3DEXPERIENCE Order Type using the mapping specified in the MB_DS_ORDER_TYPE - DS Order Type Advanced Determination. If there is 3DEXPERIENCE Order Type information in the DELMIA Apriso Database, the mapping from the Determination takes precedence.

EmployeeNo	Char	EmployeeNo of the employee performing the action that triggered the status change. EmployeeNo in outgoing json.
Facility	Char	Facility where the Order is performed. Facility in outgoing json.
OprSequenceNo	Char	Operation Number. OperationNo in outgoing json.
WorkCenter	Char	Work Center where the Order is performed. WorkCenter in outgoing json.
LastReleaseFromHold	Boolean	Specifies if the submitted release is the last one for the order. LastReleaseFromHold in outgoing json (available for the UnHold Order and UnHold Operation Message).

Create Issue

This message uses data from CAPA and CAPA_LINK tables to send details of an Issue created in DELMIA Apriso to the 3DEXPERIENCE Platform.

Field	Description
Message	► Create Issue
Refresh Inputs	Refreshes the list of available Inputs (Required and Dynamic) from the Message definition.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
TransactionTime	DateTime	Transaction Time in UTC.
CAPARecordNo	Char	Record No of the Issue. Taken from the RecordNo column in the CAPA table.

Dynamic Inputs

The Inputs can be added to the Function by selecting the check boxes on the Dynamic Inputs list.

Name	Data Type	Description
LocalTransactionTime	DateTime	Transaction Time adjusted for local time zone.

3DSwym Post Creation

This message creates a post in a 3DSwym Community on the 3DEXPERIENCE Platform.

Field	Description
Message	► Create Post
Refresh Inputs	Refreshes the list of available Inputs (Required and Dynamic) from the Message definition.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
TransactionTime	DateTime	Transaction Time in UTC.
Title	Char	Title of the 3DSwym post.
Content	Char	Content of the 3DSwym post.

Dynamic Inputs

The Inputs can be added to the Function by selecting the check boxes on the Dynamic Inputs list.

Name	Data Type	Description
LocalTransactionTime	DateTime	Transaction Time adjusted for local time zone.
Signature	Char	Signature line appended to the content of the post
CommunityID	Char	3DSwym Community unique identifier on the 3DEXPERIENCE Platform.
CommunityName	Char	Name of the 3DSwym Community on the 3DEXPERIENCE Platform.
LinkURL	Char	URL of the link included at the end of the post.
LinkTitle	Char	Title of the link included in the post.
LinkDescription	Char	Description of the link included in the post.
LinkPreviewImageURL	Char	URL of an image to be displayed with the link in the post.
AttachmentIDs	List of Integer	IDs from the ATTACHMENT Table.
DocumentIDs	List of Integer	IDs from the DOCUMENT Table.

Localization settings of the user determine which language version of the document will be included with the post.
A document will not be included if it does not have the language version corresponding to the localization settings of the user.

Get Process Header Topic

The Get Process Header topic retrieves the properties of a manufacturing process, including any enterprise attributes.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
Process3DExperienceID	Char	3DEXPERIENCE Platform ID of the process.

Import Process Topic

The Import Process topic retrieves a manufacturing process and creates equivalent entities that allow it to be used in DELMIA Apriso.

If the **Is reserved from first step** parameter of the Primary Resource on the 3DEXPERIENCE platform is set to **True**, the resources linked to the Operation are duplicated to the first Step.

Importing the process does not import the `OperationFlow` or `StepFlow` information that Process Builder needs for Processes built outside of Process Builder.
`3DX.Message.Processing.ManufacturingProcess.Released` operation can be used to import the Process, and automatically call the necessary Business Components to import the information flow. Otherwise, the `BuildOperationFlowLayout` and `BuildOperationStepsFlow` Business Components must be called manually to build the information for the Process.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
Process3DExperienceID	Char	3DEXPERIENCE Platform ID of the process.

Get Data Collect Plan Row Custom Attributes Topic

The Get Data Collect Plan Row Custom Attributes topic retrieves custom attributes from 3DEXPERIENCE platform Data Collect Plan Rows. After that, you can integrate these data with DELMIA Apriso.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
DataCollectPlanRow3DExperienceID	Char	Data Collect Plan Row ID in 3DEXPERIENCE Platform.

Get Data Collect Plan Custom Attributes Topic

The Get Data Collect Plan Custom Attributes topic retrieves custom attributes from 3DEXPERIENCE platform Data Collect Plans. After that, you can integrate these data with DELMIA Apriso.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager.

This input is only available when the From PlatformAlias Input option is selected.

DataCollectPlan3DExperienceID	Char	Data Collect Plan ID in 3DEXPERIENCE Platform.
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Get Operation Custom Attributes Topic

The Get Operation Custom Attributes topic retrieves custom attributes from 3DEXPERIENCE platform Operations. After that, you can integrate these data with DELMIA Apriso.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
Operation3DExperienceID	Char	Operation ID in 3DEXPERIENCE Platform.

Get Manufacturing Item Custom Attributes Topic

The Get Manufacturing Item Custom Attributes topic retrieves custom attributes from 3DEXPERIENCE platform Manufacturing Items. After that, you can integrate these data with DELMIA Apriso.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
ManufacturingItem3DExperienceID	Char	Manufacturing Item ID in 3DEXPERIENCE Platform.

Get Process Custom Attributes Topic

The Get Process Custom Attributes topic retrieves custom attributes from 3DEXPERIENCE platform Processes. After that, you can integrate these data with DELMIA Apriso.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
Process3DExperienceID	Char	3DEXPERIENCE Platform ID of the Process.

Get Process Organization Topic

The Get Process Organization topic retrieves an organizational entity responsible for a Process from 3DEXPERIENCE platform Processes. After that, you can filter out processes that do not apply to a DELMIA Apriso system.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
Process3DExperienceID	Char	3DEXPERIENCE Platform ID of the Process.

Get Resource Implement Links Topic

The Get Resource Implement Links topic retrieves Resources for a given Operation that use the old platform data model.

The 3DEXPERIENCE Platform may require a significant amount of time to retrieve Resource Implement Links.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.

Dynamic Inputs

The Inputs can be added to the Function by selecting the check boxes on the Dynamic Inputs list.

Name	Data Type	Description
ProcessOperationID	Int	ID of the Process Operation.
ProcessOperationStepID	Int	(Optional input) ID of the Process Operation Step. Used to specify retrieving resources at an Operation level.
WIPOrderNo	Char	WIP Order Number.
WIPOrderType	Int	WIP Order Type.
OprSequenceNo	Char	Operation Sequence Number.
StepSequenceNo	Int	(Optional input) Step Sequence Number. Used to specify retrieving resources at a Step level.

Get Resource Custom Attributes Topic

The Get Resource Custom Attributes topic retrieves custom attributes from 3DEXPERIENCE platform Resources. After that, you can integrate these data with DELMIA Apriso.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
Resource3DExperienceID	Char	Resource ID in 3DEXPERIENCE Platform.

Import Resource Topic

The Import Resource topic retrieves Resource information from 3DEXPERIENCE platform and creates equivalent entities that can be used in DELMIA Apriso.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
PlatformAlias	Char	Alias of the platform configured in Configuration Manager. This input is only available when the From PlatformAlias Input option is selected.
Resource3DExperienceID	Char	Resource ID in 3DEXPERIENCE Platform.

The function returns "-1" as ResourceID output if the Resource that the function tried to import was skipped or ignored.

Import IP Protection and Export Control Topic

The Import IP Protection and Export Control topic is used to link a classification to an object.

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
DSID	Char	Physical ID of the platform object. Consists of 36 hexadecimal characters.

Type	Char	Type of the platform object. Supported types: ▶ Alert ▶ Checklist ▶ DataCollectPlan ▶ Document ▶ MfgItem ▶ Operation ▶ Process ▶ Resource ▶ SignOff ▶ Instruction
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Get Translation

The Get Translation topic retrieves the latest translations of the 3DEXPERIENCE entities from the 3DEXPERIENCE platform. If the **LanguageIDs** dynamic input is not used, the translations are retrieved for the list of languages configured for the Platform Alias in the Configuration Manager. For details, see [Integration Options](#) .

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
DSIDs	List of Char	Physical IDs of the platform objects. Each ID consists of 36 hexadecimal numbers.

Dynamic Inputs

The Inputs can be added to the Function by selecting the check boxes on the Dynamic Inputs list.

Name	Data Type	Description
LanguageIDs	List of Integer	A list of IDs of the languages for which translations are retrieved.

Import Translation

The Import Translation topic retrieves the translations from the 3DEXPERIENCE platform for a DELMIA Apriso object and any of its child objects. The list of languages for which translations are provided is configured in the Configuration Manager. For details, see [Integration Options](#) .

Required Inputs

The values of these Inputs have to be filled and valid to use the Function.

Name	Data Type	Description
DSID	Char	Physical ID of the platform object. Consists of 36 hexadecimal characters.
Type	Char	Type of the platform object. Supported types: ▶ Alert ▶ Checklist ▶ DataCollectPlan

- ▶ Order
- ▶ MfgItem
- ▶ Operation
- ▶ Process
- ▶ Resource
- ▶ SignOff
- ▶ Instruction